Project Initialization and Planning Phase

Date	1 July 2026
Team ID	XXXXXX
Project Title	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution introduces an **AI-powered Emotion Detection and Learning Support Engine (EDLSE)** that identifies a learner's emotional state using facial expressions, voice, and text analysis. The system leverages deep learning models to detect emotions in real time and provides personalized learning recommendations, emotional insights, and progress reports to improve engagement and academic performance.

Project Overview

Topic	Description
Objective	Develop an intelligent application that detects learners' emotions using AI and provides personalized learning support to enhance engagement, motivation, and academic performance.
Scope	The system supports webcam/image upload, voice and text input, emotion detection, personalized learning recommendations, progress tracking, report generation, and interactive dashboards. .

Problem Statement

Topic Description

Description Traditional learning platforms evaluate academic performance but fail to

recognize learners' emotional states such as stress, confusion, boredom, or frustration, which significantly affect learning outcomes.

Impact An AI-powered emotion-aware learning system improves learner engagement, identifies emotional challenges early, provides personalized learning assistance, and enhances the overall educational experience.

Proposed Solution

Topic	Details
Approach	Use Computer Vision, Speech Processing, Natural Language Processing (NLP), and Deep Learning models (CNN, LSTM, Transformers) to detect emotions and recommend personalized learning resources
Key Features	◆ Real-time facial emotion ◆ detection ◆ • Voice emotion recognition ◆ • Text sentiment analysis ◆ • Emotion classification ◆ (Happy, Sad, Angry, Neutral, ◆ Confused, etc.) • Personalized learning recommendations • Interactive dashboard

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	**Processor (CPU)**	Intel Core i5 / AMD Ryzen 5 or higher
	Memory (RAM)	8 GB or above
	Storage	10 GB free SSD space
	Camera & Microphone	HD Webcam and Microphone for real-time emotion detection (optional but recommended)
Software	**Application Framework**	Streamlit
	Programming Language	Python 3.10+
	AI/ML Libraries	TensorFlow, Keras, Scikit-learn
	Computer Vision	OpenCV, MediaPipe
	NLP & Speech Processing	NLTK, TextBlob, Transformers, Librosa

Resource Type	Description	Specification / Allocation
	Data Processing	NumPy, Pandas
	Visualization & Reporting	Matplotlib, Plotly, FPDF (or ReportLab)
	Development Environment	Visual Studio Code
Data	**Dataset**	FER-2013 (Facial Emotion Recognition), RAVDESS (Speech Emotion Recognition), Emotion-labelled text datasets, and user interaction logs